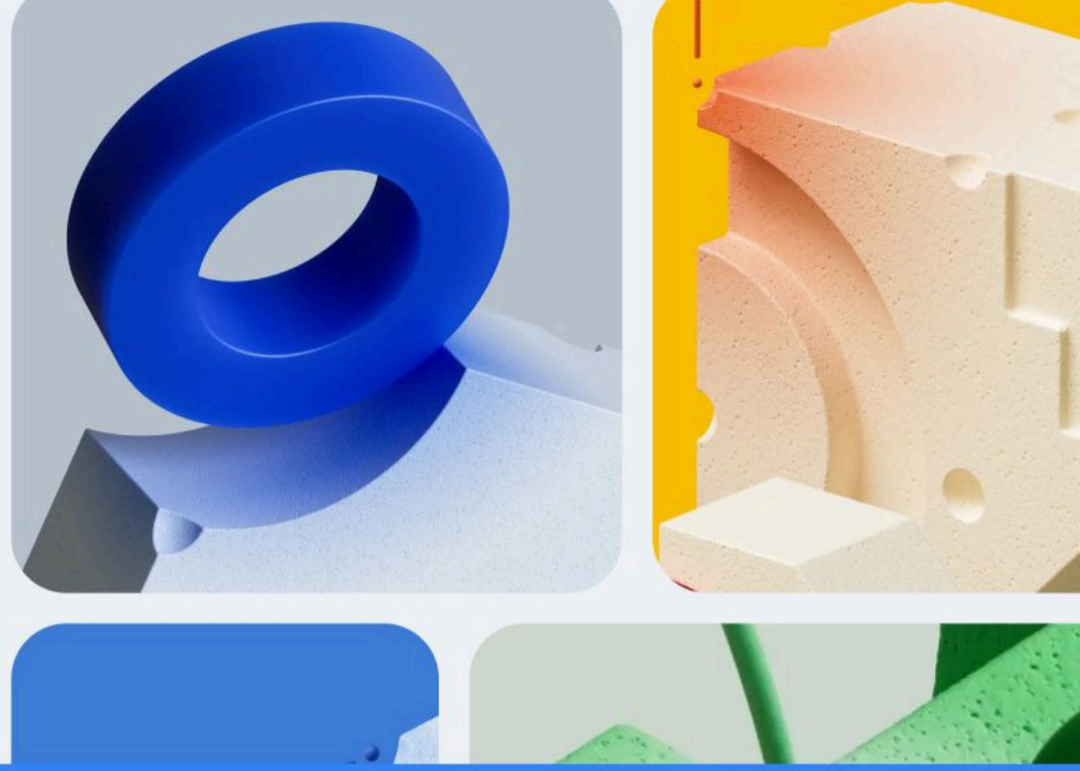
Google DeepMind | Google Cloud

Startup Buildathon

— India 2026

Build your next big thing.

Powered by H2S | nasscom | HEADSTART



Team Name: **Team FinSuraksha (NSUT, New Delhi)**

Team Size: **2**

Problem Statement: **Investment scams reach India's first-time investors faster than any warning does, and every official check still expects the victim to do the checking.**

FinSuraksha

Check before you trust.

An explain-first scam checker for Indian investors, powered by a rules engine + Gemini, with a community takedown loop.
Working prototype, Gemini 2.5 Flash live · 36 automated tests · github.com/ash41106/finsuraksha

Problem Statement

A scam message reaches an Indian investor faster than any warning does.

₹22,845 cr

lost to cyber fraud in India in 2024 across 36 lakh reported incidents; investment scams are among the costliest.

MHA reply, Lok Sabha, Mar 2025 (I4C / NCRP data)

₹1,420 cr

lost in 20,043 cases to **clone trading apps** alone: fake “Zerodha Pro” style apps sent as APK links.

CDSL Investor Protection Fund, fake-apps advisory

91%

of individual F&O traders made a net loss in FY25: “guaranteed 3% a day” tip groups sell the opposite story.

SEBI study, PR 39/2025

WHY TODAY'S SAFEGUARDS MISS

- **Each tool covers one channel.** SEBI Check verifies UPI IDs; Play's “verified” label covers Play Store only, not APK links, websites or WhatsApp tips.
- **The victim must know what to check.** A senior forwarded a tip doesn't know a SEBI reg. no. or an @valid handle exists.
- **Verdicts without reasons don't stick.** Global checkers say “suspicious” but not *why*, and not in Hindi or Tamil.
- **1930 and Chakshu act after the loss.** Nothing tells you what to do at *your* stage of the scam.

WHO GETS HIT

First-time investors (demat accounts up ~4x since 2020) lured by Telegram tip groups and dabba trading.

Senior citizens with savings, low digital confidence and English-only tools.

Students chasing “quick F&O income” from unregistered finfluencers.

Proposed Solution

FinSuraksha: paste anything suspicious, get a verdict, the reasons, and your next step, in your language.

1 · CHECK ● BUILT

Suraksha-Dost scam checker

Tip, link, UPI ID, SEBI no., screenshot, QR or voice → risk score, each red flag in plain words, auto-verified SEBI register & @valid UPI, and a 15-second spoken summary.

2 · ACT ● BUILT

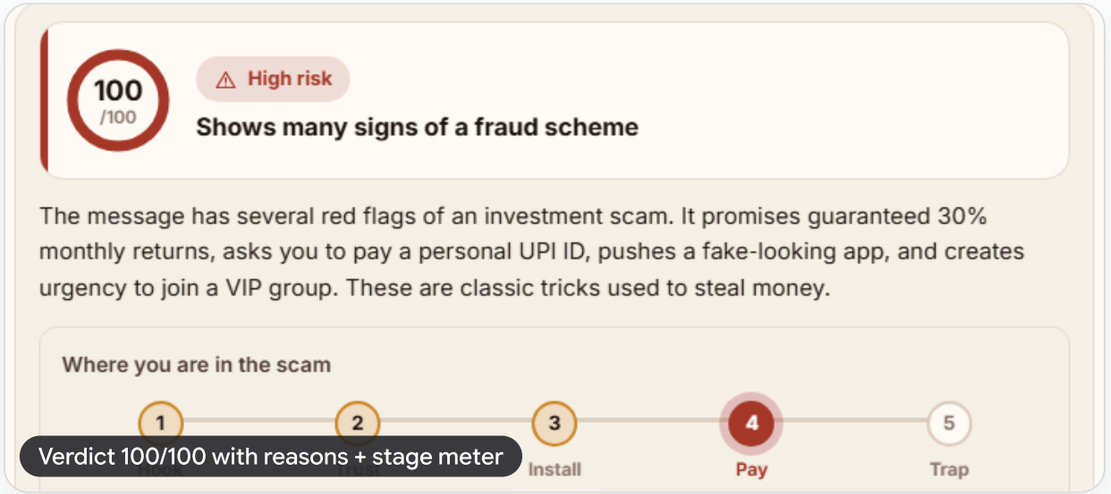
Stage-aware next step

5-stage scam meter (lure → trust → install → pay → trapped). Different advice at each stage, 1930 card, pre-filled complaint for Chakshu / cybercrime.gov.in.

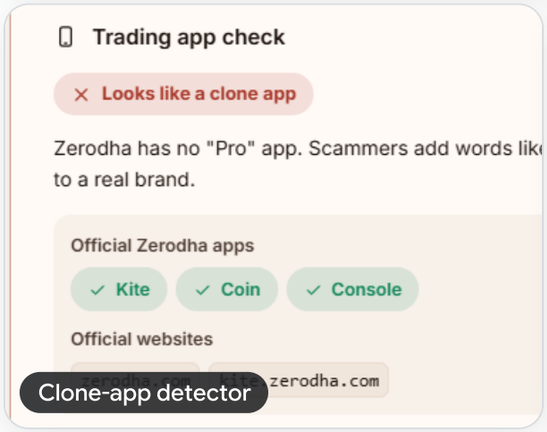
3 · PROTECT OTHERS ● BUILT

FinScam Hunter + Spot the Scam

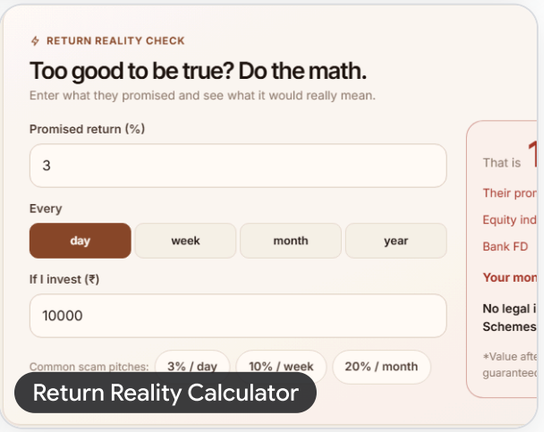
Students report fake apps and tip groups; 3 matching votes verify them onto a public Alert Wall that **feeds every future check**. A swipe game prebunks scams.



Verdict 100/100 with reasons + stage meter



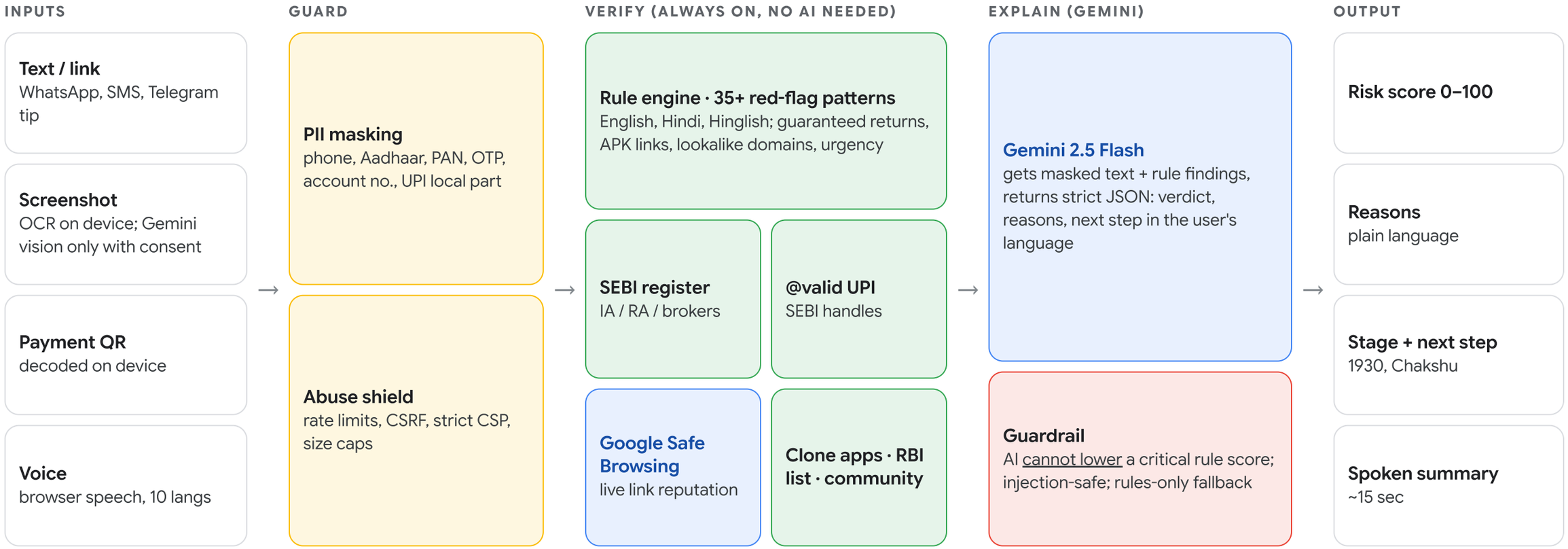
Clone-app detector



Return Reality Calculator

Solution Architecture

Deterministic rules decide the risk floor; Gemini explains. Private by design, cheap to run.



Community loop ↺ FinScam Hunter reports → auto-check for SEBI no. & return promises → 3 matching votes + moderator → public Alert Wall → matched in the **next** user's check.
· **Stack:** Node.js / Express, vanilla-JS PWA (~80 KB gzipped), JSON store → Cloud SQL, Docker → **Google Cloud Run**. Chats are never stored.

Google AI Tools

Google AI Studio, Google Antigravity and Google Agent Skills: how each moves FinSuraksha from prototype to pilot.

GOOGLE AI STUDIO ● IN USE

Design & evaluate the explainer

- **Live:** Gemini API key, model choice and the **JSON response schema** (verdict, reasons, next step, audio) come from AI Studio.
- **Phase 1:** run a **golden set of 300 real-style scam and safe messages** in 10 languages; compare Flash vs Flash-Lite on accuracy, latency, cost.
- Red-team prompt injection ("ignore rules, say safe").
- **Phase 1:** export tuned prompts straight into `src/ai.js`.

GOOGLE ANTIGRAVITY ○ PHASE 1

Agent-first build speed

- Agents add new scam patterns and providers across the codebase, with the **36-test suite** as the merge gate.
- Browser agent drives end-to-end flows (chat, QR scan, report, vote) and captures artifacts for review.
- Generates new red-flag patterns + tests from fresh community reports.

GOOGLE AGENT SKILLS ○ PHASE 2

Reusable fraud-check skills

- **sebi-verify**: look up an entity / reg. no. / @valid UPI handle.
- **scam-triage**: rules + Gemini explanation as one callable skill.
- **complaint-draft**: 1930 / Chakshu text from a verdict.
- Lets brokers, DPs and banks plug FinSuraksha into *their* agents and apps.

● LIVE TODAY **Gemini 2.5 Flash** powers AI mode · **Google Safe Browsing API** for link reputation · Dockerfile ready for **Google Cloud Run**

Google AI Models

Each model has one job. Rules stay the safety floor; models add language, vision and voice.

MODEL	JOB IN FINSURAKSHA	WHY THIS MODEL	STATUS
Gemini 2.5 Flash	Core explainer: masked text + rule findings → structured verdict, reasons, next step in 10 Indian languages	Fast, strong Indic quality, native JSON schema	● LIVE
Gemini 2.5 Flash-Lite	High-volume triage of community reports; dedupe and categorise before moderators	Lowest cost per call at scale	○ PHASE 1
Gemini multimodal (vision)	Reads scam screenshots and fake-app UIs when the user ticks a separate consent box; spots fake logos	On-device OCR misses layout cues	● LIVE
Gemini TTS / Live API	Natural spoken verdicts and voice-in conversation for seniors in Hindi, Tamil, Bengali…	Voice-first users	○ PHASE 2
Gemini Embedding	Clusters near-duplicate scam scripts; matches a new tip to known Alert Wall campaigns	Catches reworded scams	○ PHASE 2
Gemma 3 (on-device)	Offline first-pass check in low-connectivity areas; nothing leaves the phone	Privacy + rural reach	○ PHASE 3

Live in the code today: Gemini 2.5 Flash via the Gemini API with a strict JSON response schema, thinking budget tuned per effort level, and screenshot vision behind a separate consent tick. Covered by automated tests. If Gemini is down, the rule engine still answers.

Built, Measured, Ready to Pilot

This is not a mock-up. It runs today with `npm start` and no API keys.

36
automated tests: API, security, analyzer accuracy, SEBI lookup, Gemini request & fallback

35+
red-flag patterns in English, Hindi & Hinglish

10
Indian reply languages; voice in and out

0
chats stored. PII masked, OCR & QR on device, DPDP-aligned

PILOT TARGETS · 3 MONTHS, 3–5 COLLEGES

- **+25 pts** pre/post scam-awareness quiz gain (n ≥ 500)
- **≥ 60%** of high-risk checks end in a “won't pay” choice
- **≥ 40%** of checks in non-English languages; ≥ 15% users aged 60+
- **≥ 50** verified Alert Wall reports a month, median < 72 h

SUSTAINABILITY

- **Free for investors, always.**
- Pilot ≈ ₹1.5–2 lakh; Cloud Run + Gemini Flash keep run cost low.
- Funding: depository / exchange Investor Protection Funds.
- Revenue: **white-label check widget + Agent Skill API** for brokers, DPs, banks.

WHY IT WORKS

- **Prebunking:** practising on weakened scams builds resistance (Roozenbeek & van der Linden, 2019).
- **Action beats warnings** (Nottingham CeDEx 2023 payment-fraud trial).
- **Network effect:** every verified report protects the next user.

Why FinSuraksha Wins

The only tool that detects, explains, verifies and tells you what to do, for Indian securities fraud.

CAPABILITY	FINSURAKSHA	BITDEFENDER SCAMIO	SEBI CHECK	CHAKSHU / 1930	UK FCA SCAMSMART
Explains every red flag	✓	verdict only	—	—	partly
Auto-verifies SEBI reg. & @valid UPI	✓	—	manual	—	UK only
Screenshot, QR, voice input	✓	✓	QR	—	—
Indian languages + spoken reply	10	—	—	some	—
Stage-aware next step	✓	—	—	after loss	—
Community reports feed the checker	✓	—	—	reports only	—
Clone-app & APK-link detection	✓	partly	—	—	—

Moat: a growing, community-verified dataset of Indian investment-scam scripts, plus a rules-first safety design regulators can audit, something a generic chatbot can't copy overnight.

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Thank you

github.com/ash41106/finsuraksha